Graphic Era HILL UNIVERSITY
Established by an Act of the State Legislature of Uttarakhand (Adhiniyam Sankhya 12 of 2011)
University under section 2(f) of UGC Act, 1956

End Term (Odd) Semester Examination December 2024

Roll no.....

Name of the Course and semester: **B.Tech ECE VII**
Name of the Paper: **Neural Network and Machine Learning**
Paper Code: **TOE 701**
Time: 3 hour Maximum Marks: 100

Note:
(i) All the questions are compulsory.
(ii) Answer any two sub questions from a, b and c in each main question.
(iii) Total marks for each question is 20 (twenty).
(iv) Each sub-question carries 10 marks.

Q1. (2X10=20 Marks)
a. Explain how a neural network adjusts its weights during learning using the error function, backpropagation, and gradient descent to minimize errors and improve performance. CO1
b. Illustrate the structure and working of a computational neuron model, explaining its components like inputs, weights, activation function, and output, and how they interact to process and transmit information. CO1
c. Compare supervised, unsupervised, and reinforcement learning, focusing on their learning approaches, data needs, and examples. CO1

Q2. (2X10=20 Marks)
a. Describe pattern classification in multilayer feedforward neural networks, focusing on data flow, weight adjustment, and how the network learns to classify input patterns. CO2
b. Explain in detail the perceptron, including its structure, working mechanism, and how it is utilized to classify patterns effectively in machine learning. CO2
c. Outline the architecture of a multilayer feedforward neural network, describing its components, layer connections, and role in processing and learning complex data patterns. CO2

Q3. (2X10=20 Marks)
a. In what way does statistical learning theory provide a framework for evaluating the capacity of machine learning models to learn from data and make accurate predictions on unseen examples? CO3
b. In what ways does the application of regularization theory enhance the effectiveness of Radial Basis Function (RBF) networks, particularly in handling complex data patterns and preventing overfitting during training? CO3
c. How do Radial Basis Function (RBF) networks approximate complex functions, and what role do their hidden units play in this process? CO3

Q4. (2X10=20 Marks)
a. In what manner does a Self-Organizing Map (SOM) facilitate the clustering of patterns based on their inherent characteristics? CO4
b. How does Kohonen's Self-Organizing Map operate to organize data in unsupervised learning tasks? CO4
c. How do Counter Propagation Networks assist in recognizing and classifying patterns? CO4

Q5. (2X10=20 Marks)
a. How do neural networks store and retrieve patterns during their operation? CO5
b. How does the Hopfield model function, and what role does it play in storing patterns? CO5
c. How are neural networks utilized to solve practical problems in real-world case studies? CO5